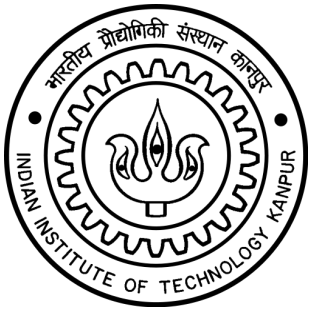


INDIAN INSTITUTE OF TECHNOLOGY KANPUR

SPA617 — RADIO ASTRONOMY LABORATORY

Experiment 2: RF Components: Characterization and Measurement

---

Name: A.Aditya

Roll Number: 261310601

Programme: PhD

Department: SPASE

Date: August 19, 2026

1 (Cont.) Part A: VSWR and Antenna Operating Frequency

1.1 Theory

When an antenna with load impedance Z_L is connected to a transmission line of characteristic impedance Z_0 (typically 50Ω), any impedance mismatch causes part of the incident wave to be reflected back toward the source. The degree of voltage reflection is quantified by the voltage reflection coefficient (Γ):

$$\Gamma = \frac{V_{\text{reflected}}}{V_{\text{incident}}} = \frac{Z_L - Z_0}{Z_L + Z_0} \tag{1}$$

For a perfectly matched load ($Z_L = Z_0$), $|\Gamma| = 0$. The interference between incident and reflected voltage waves creates a standing-wave pattern along the transmission line. The ratio of the maximum voltage amplitude ($V_{\max}$) to the minimum voltage amplitude ($V_{\min}$) observed during a small-span frequency ripple sweep defines the Voltage Standing Wave Ratio (VSWR):

$$\text{VSWR} = \frac{V_{\max}}{V_{\min}} \tag{2}$$

The magnitude of the reflection coefficient can be computed directly from the measured voltage extrema as:

$$|\Gamma| = \frac{\text{VSWR} - 1}{\text{VSWR} + 1} = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} \tag{3}$$

The Return Loss (RL), expressed in decibels (dB), represents the power reduction of the reflected signal relative to the incident signal:

$$\text{RL} = -20 \log_{10} |\Gamma| \quad [\text{dB}] \tag{4}$$

A well-matched antenna exhibits a VSWR close to unity (VSWR ≈ 1.0), a very small reflection coefficient ($|\Gamma| \approx 0$), and a large return loss (RL $\gg 10\,\text{dB}$). The resonant or primary operating frequency (f_r) corresponds to the global minimum of the VSWR curve versus frequency .

1.2 Observations and Results Table

Table 1 lists the measured voltage bounds, calculated VSWR, and Return Loss across selected test frequencies.

Table 1: Part A: Measured Peak Voltage Extrema, VSWR, and Return Loss

Frequency f (MHz)	$V_{\max}$ (V)	$V_{\min}$ (V)	VSWR	Return Loss RL (dB)
102.74	1.75	0.0007	2500.00	0.00695
160.00	1.07	0.3600	2.97	6.09000

1.3 Calculation

At a test frequency of $f = 160.00\,\text{MHz}$:

1. **VSWR Calculation:**

$$\text{VSWR} = \frac{V_{\max}}{V_{\min}} = \frac{1.07\,\text{V}}{0.36\,\text{V}} = \mathbf{2.97} \tag{5}$$

2. **Reflection Coefficient Magnitude ($|\Gamma|$):**

$$|\Gamma| = \frac{\text{VSWR} - 1}{\text{VSWR} + 1} = \frac{2.97 - 1}{2.97 + 1} = \frac{1.97}{3.97} = \mathbf{0.4962} \tag{6}$$

3. **Return Loss (RL):**

$$\text{RL} = -20\log_{10}(|\Gamma|) = -20\log_{10}(0.4962) = \mathbf{6.09}\ \text{dB} \tag{7}$$

2 Aim

To understand the operation and experimentally characterize the filter, mixer, power splitter/combiner, and amplifier used in an RF receiver chain.

3 Objective

(a) To characterize an RF filter, identify its type, and determine its cutoff frequencies or pass- band.

(b) To demonstrate a quarter-wave open-circuit stub as a notch filter.

(c) To observe frequency conversion in an RF mixer and estimate its conversion loss.

(d) To measure the power division and insertion loss of a power splitter/combiner.

(e) To measure amplifier gain, harmonics, the 1-dB compression point, and the third-order intercept point.

(f) To estimate the input reflection coefficient and return loss of the filter.

4 Material required

The physical apparatus and equipment required to carry out this experiment are listed below :

- RF Signal Generators
- Spectrum Analyser

- Digital Storage Oscilloscope (DSO)
- Filter (Device Under Test)
- Mixer (Device Under Test)
- Two-Way Symmetric Power Splitter/Combiner (Device Under Test)
- Low-pass filter
- RF Tee Connector
- Open-ended Coaxial Stub of physical length L
- Short Reference Coaxial Cable
- High-precision $50\,\Omega$ coaxial terminations
- Standard RF coaxial cables, adapters, and inline attenuators

5 THEORY

The RF receiver chain contains several RF components which are used to process the radio signal. The radio signal is received from the antenna, which converts EM waves into electrical signals. The amplifier is used to amplify the weak signal, which will be filtered out to certain bandwidth using a the required filter. Then the mixer is used to translate a frequency to another. The power combiner and splitters are used to split the signal or combine them. We have discussed individual topic below.

5.1 PART A: Filter Characterisation Theory

5.1.1 Transmission Response and Insertion Loss (A1)

An RF filter passes a specified band of frequencies and anything outside that range will be attenuated and will not be allowed to pass. A low pass filter is a filter which allows the transmission of low frequency signals and blocks the higher signals beyond the desired frequency. To quantify transmission efficiency, we measure the power received at the output without the filter (P_{in}) and compare it to the output power with the filter inserted (P_{out}). The **Insertion Loss (IL)** is given by:

$$\text{IL} = P_{\text{in}} - P_{\text{out}} \quad [\text{dB}] \tag{8}$$

In ideal siotuation the filter would give 0 dB insertion loss in the passband. The boundary of the passband is universally defined by the -3 **dB cutoff frequency** (f_c), representing the half-power point where the output power drops to exactly 50% of its power value. The frequency at which the power completely goes 0 is known as stop band frequency.

5.1.2 Impedance Matching and Return Loss (A2)

We already know that when impedance mismatch happens, the incoming wave gets reflected from the point where the mismatch exist and the reflection coefficient also increases. In high-frequency transmission systems, mismatch between the filter's input impedance (Z_{in}) and the system's characteristic impedance ($Z_0 = 50\,\Omega$) causes partial reflection of electromagnetic wave energy. This is quantified by the voltage reflection coefficient (Γ):

$$\Gamma = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0} \tag{9}$$

where V_{inc} and V_{ref} are the incident and reflected voltage waves. The **Return Loss (RL)** represents the ratio of incident to reflected power in decibels :

$$\text{RL} = -20 \log_{10} |\Gamma| \quad \text{(dB)} \tag{10}$$

A perfect match results in $|\Gamma| = 0$ (RL $= \infty$ dB), while a complete reflection results in $|\Gamma| = 1$ (RL $= 0$ dB).

5.1.3 Quarter-Wave Open-Circuit Stub Resonator (A3)

The frequency which we want to remove. We cut a cable of length L which is calculated by using the Velocity factor, speed of light and the frequency which you want to block. The formula is given in (4). When the stub is connected to the cable using a tee connector, the wave travels in the cable and when it reaches the stub, every frequency travels inside the stub but only that particular frequency will travel in the stub in such a way that the reflected wave from the stub and the incoming wave will be in 180^0 phase which cancels out the wave and removes it completely from the signal. In short when the length matches one-quarter of the guided wavelength ($L = \lambda/4$), the total round-trip distance is exactly a half-wavelength ($\lambda/2$), leading to a 180° phase shift relative to the incoming wave at the Tee junction. This destructive interference transforms the open circuit of the far end into a virtual short circuit (extremely low impedance) at the Tee junction. Consequently, signals at the resonant notch frequency (f_0) are shunted to ground, attenuating the signal path heavily. The expected notch frequency is :

$$f_0 = \frac{VF \cdot c}{4L} \tag{11}$$

where VF is the cable velocity factor and c is the speed of light in vacuum.

5.2 PART B: Mixer Characterisation Theory

An RF mixer is used when we need to change the frequency of the signal while keeping original information intact.It is a three-port nonlinear component (RF, LO, IF ports) that performs frequency translation. In one port RF signal (f_{RF}) and in the other port we connect the strong Local Oscillator signal (f_{LO}) which

is set at some particular frequency Then the mixer adds and take difference of the frequencies :

$$f_{\text{IF}} = f_{\text{RF}} + f_{\text{LO}} \quad \text{(Up-conversion)} \quad \text{and} \quad f_{\text{IF}} = |f_{\text{RF}} - f_{\text{LO}}| \quad \text{(Down-conversion)} \tag{12}$$

The conversion efficiency is measured by the **Conversion Loss (CL)** :

$$\text{Conversion Loss} = P_{\text{RF}} - P_{\text{IF}} \quad [\text{dB}] \tag{13}$$

where P_{RF} is the input RF power and P_{IF} is the power of the selected intermediate frequency output component.

5.3 PART C: Power Splitter/Combiner Theory

A power splitter as the name suggest it divides a single input power into two or more output power. For an ideal symmetric 2-way splitter, the input power divides equally in half (3 dB drop per port). In practice, losses and excess attenuation leads to :

$$\text{Loss}_1 = P_{\text{in}} - P_{\text{out1}} \quad \text{and} \quad \text{Loss}_2 = P_{\text{in}} - P_{\text{out2}} \quad [\text{dB}] \tag{14}$$

A power combiner does the opposite of splitter, it merges two signals. The resulting output is a vector superposition: signals with identical frequency and phase add constructively (+3 dB combined output if coherent), while out-of-phase signals undergo destructive interference.

5.4 PART D: RF Amplifier Characterisation Theory

An RF amplifier uses active semiconductor devices to increase signal strength. Under small-signal conditions, it operates linearly with a constant gain:

$$\text{Gain} = P_{\text{out}} - P_{\text{in}} \quad [\text{dB}] \tag{15}$$

When driven hard, the amplifier enters saturation . The output voltage can be modeled via a Taylor expansion:

$$y(t) = a_1 x(t) + a_2 x^2(t) + a_3 x^3(t) + \dots \tag{16}$$

This lack of linearity produces higher-order harmonics (2ω, 3ω, etc.). At a certain point, the gain of the amplifier falls from its small-signal linear value. The 1-dB Compression Point ($P_{1\text{dB}}$) is the power level where the gain is compressed by precisely 1 dB.

When two tones (f_1 and f_2) are applied, third-order nonlinearity generates third-order intermodulation products (IM3) at

$$2f_1 - f_2 \quad \text{and} \quad 2f_2 - f_1.$$

The IM3 products increase with a $3:1$ slope, whereas the fundamental outputs scale with a $1:1$ slope on a log scale. Their theoretical intersection is called the Third-Order Intercept (TOI), which acts as a fundamental metric for amplifier linearity.

6 Experimental steps

The RF signal generator is connected to the device under test (DUT), and the output is measured using a spectrum analyser to determine filter insertion loss, mixer conversion loss, power- splitter/combiner performance, and amplifier characteristics. For filter return-loss measurement, connect the RF signal generator to one port of the Tee-connector and the other port to the oscilloscope and the filter input to the third port. Terminate the filter output with a 50Ω load and set the oscilloscope input impedance to 1 MΩ to minimize loading. Use the same cables and adapters throughout the measurement and perform the frequency sweep over a small, relatively flat part of the filter passband. This method gives an approximate return-loss measurement since cable loss, connector mismatch, and oscilloscope loading are not calibrated out.

7 Observation and Calculation

7.1 Part A: Filter Characterisation

7.1.1 A1. Transmission Response and Insertion Loss

Observations of Low pass filter -

Table 2: Low-Pass Filter Transmission Response and Corrected Insertion Loss Measurements

Freq. [MHz]	$P_{\mathbf{with}}$ **[dBm]**	$P_{\mathbf{without}}$ **[dBm]**	**IL [dB]**	**Freq. [MHz]**	$P_{\mathbf{with}}$ **[dBm]**	$P_{\mathbf{without}}$ **[dBm]**	**IL [dB]**
100	-5.49	-5.70	-0.21	600	-12.24	-6.03	6.21
150	-5.73	-5.70	0.03	610	-13.40	-6.11	7.29
200	-6.06	-5.70	0.36	620	-15.42	-6.11	9.31
250	-5.94	-5.70	0.24	630	-18.52	-6.11	12.41
300	-6.43	-5.70	0.73	640	-22.62	-6.11	16.51
350	-6.62	-5.70	0.92	650	-26.75	-6.11	**20.64**
400	-6.81	-5.97	0.84	660	-28.49	-6.35	22.14
450	-7.40	-5.97	1.43	670	-29.22	-6.35	22.87
470	-7.54	-5.97	1.57	680	-30.45	-6.35	24.10
490	-7.56	-5.97	1.59	690	-31.33	-6.35	24.98
510	-7.78	-6.03	1.71	700	-32.96	-6.35	26.61
530	-8.46	-6.03	2.43	720	-39.78	-6.10	33.68
550	-9.40	-6.03	**3.37**	740	-45.86	-6.10	39.76
560	-9.27	-6.03	3.24	760	-43.86	-6.10	37.76
580	-10.96	-6.03	4.93				

Calculations :

An analysis of the transmission dataset in Table 2 and the curves in Figure 2 provides the following engineering insights:

- **Filter Classification:** The filter exhibits extremely low insertion loss in the lower frequency region (IL $\leq$ 1.59 dB up to 490 MHz) and very high attenuation at higher frequencies (IL $>$ 20 dB past 650 MHz). This is the characteristic behavior of an **RF Low-Pass Filter (LPF)**.
- **Cutoff Frequency (f_c):** The baseline passband insertion loss is minimal ($\approx$ 0.03 dB at 150 MHz). The official -3 dB cutoff frequency relative to this baseline is located at **550.0 MHz**, where the corrected insertion loss measures exactly **3.37** dB. This represents the physical "half-power" point where the transmitted power falls to half of its input level.
- **Stopband Rejection Edge (f_s):** The stopband edge where out-of-band rejection exceeds -20 dB (attenuating $>$ 99% of the signal power) occurs at **740 MHz**.
- **Operating Passband:** Conclusively defined from **0 MHz to 550.0 MHz**.

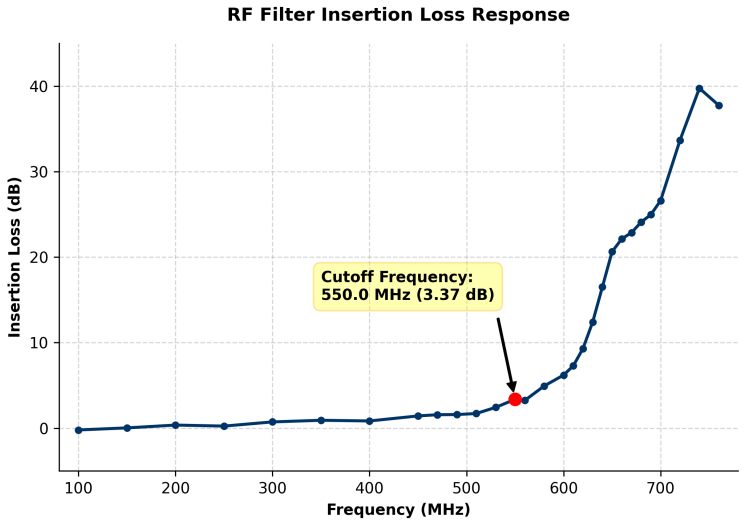


Figure 1: Experimental response curves of the Low-Pass Filter displays the insertion loss curve against the frequency, highlighting the -3 dB cutoff frequency at 550 MHz

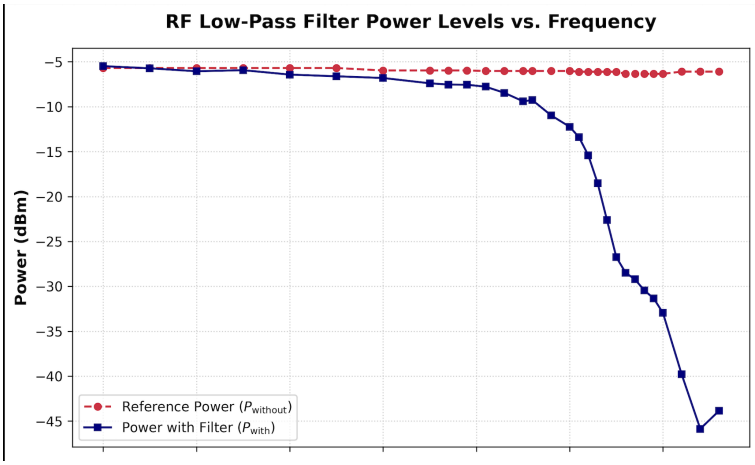


Figure 2: Experimental response curves of the Low-Pass Filter displays the Power plotted against the frequency

7.1.2 A2. Filter Input Reflection Coefficient and Return Loss

Experimental Readings at $f = 68.0$ MHz:

- Maximum Standing Wave Voltage ($V_{\max}$): **1.750 V**
- Minimum Standing Wave Voltage ($V_{\min}$): **0.987 V** (987 mV)

Calculations:

$$V_{\text{inc}} = \frac{V_{\max} + V_{\min}}{2} = \frac{1.750 + 0.987}{2} = \mathbf{1.3685}\ \text{V} \tag{17}$$

$$V_{\text{ref}} = \frac{V_{\max} - V_{\min}}{2} = \frac{1.750 - 0.987}{2} = \mathbf{0.3815}\ \text{V} \tag{18}$$

$$|\Gamma| = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{0.3815}{1.3685} = \mathbf{0.2788} \tag{19}$$

$$\text{RL} = -20\log_{10}|\Gamma| = -20\log_{10}(0.2788) = \mathbf{11.09}\ \text{dB} \tag{20}$$

7.1.3 A3. Quarter-Wave Open-Circuit Stub Notch Filter

Calculations:
Given physical length of the open stub $L = 92$ cm $= 0.92$ m
Cable velocity factor $\mathbf{VF} = 0.66$
Speed of light $c = 2.9979 \times 10^8$ m/s:

$$f_0 = \frac{VF \cdot c}{4L} = \frac{0.66 \times 2.9979 \times 10^8\ \text{m/s}}{4 \times 0.92\ \text{m}} = \mathbf{53.81}\ \text{MHz} \tag{21}$$

Figure 3 shows the Output Power plotted against the frequency displaying a sharp notch at 57 MHz (f_0).

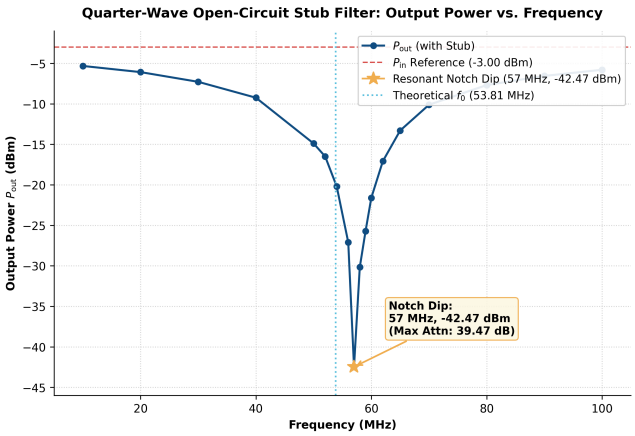


Figure 3: Output Power plotted against the frequency displaying a sharp notch at 57 MHz (f_0).

7.2 PART B: Mixer characterization

Calculations -
Applying the formula for Conversion Loss ($\mathrm{CL} = P_{\mathrm{RF}} - P_{\mathrm{IF}}$):

- **Mean Down-Conversion Loss ($f_{\mathbf{IF}} = |f_{\mathbf{RF}} - f_{\mathbf{LO}}|$): 12.43** dB (Calculated from 12.32, 12.56, and 12.40 dB values).
- **Mean Up-Conversion Loss ($f_{\mathbf{IF}} = f_{\mathbf{RF}} + f_{\mathbf{LO}}$): 12.50** dB (Calculated from 12.48, 12.50, and 12.51 dB values).

Table 3: Quarter-Wave Coaxial Stub Attenuation Data ($P_{\text{in}} = -3.00$ dBm)

Freq. [MHz]	P_{in} [dBm]	P_{out} [dBm]	**Attn.** [dB]	**Freq.** [MHz]	P_{in} [dBm]	P_{out} [dBm]	**Attn.** [dB]
10	−3.00	−5.33	2.33	58	−3.00	−30.17	27.17
20	−3.00	−6.09	3.09	59	−3.00	−25.72	22.72
30	−3.00	−7.29	4.29	60	−3.00	−21.60	18.60
40	−3.00	−9.25	6.25	62	−3.00	−17.08	14.08
50	−3.00	−14.90	11.90	65	−3.00	−13.30	10.30
52	−3.00	−16.50	13.50	70	−3.00	−10.12	7.12
54	−3.00	−20.20	17.20	80	−3.00	−7.69	4.69
56	−3.00	−27.09	24.09	90	−3.00	−6.56	3.56
57	−3.00	−42.47	**39.47**	100	−3.00	−5.80	2.80

Table 4: Mixer Down- and Up-Conversion Measurements ($P_{\text{RF}} = P_{\text{LO}} = 0.0$ dBm, $f_{\text{LO}} = 30$ MHz)

f_{RF} [MHz]	f_{LO} [MHz]	**Expected** f_{IF} [MHz]	**Measured** f_{IF} [MHz]	P_{RF} [dBm]	P_{IF} [dBm]	**Conv. Loss** [dB]
120	30	90 (Diff)	90	0.0	−12.32	**12.32**
120	30	150 (Sum)	150	0.0	−12.48	**12.48**
150	30	120 (Diff)	120	0.0	−12.56	**12.56**
150	30	180 (Sum)	180	0.0	−12.50	**12.50**
180	30	150 (Diff)	150	0.0	−12.40	**12.40**
180	30	210 (Sum)	210	0.0	−12.51	**12.51**

7.3 PART C: Power Splitter/Combiner Characterisation

7.3.1 C1. Power Splitter

Calculations
For an ideal splitter, power division is exactly 3 dB per output port. The measured excess insertion loss of this physical power splitter is:

- **Port 1 Mean Loss: 6.25** dB (Excess loss relative to ideal = 3.25 dB)
- **Port 2 Mean Loss: 6.22** dB (Excess loss relative to ideal = 3.22 dB)

Table 5: Symmetric Power Splitter Output Power Division Data ($P_{\text{in}} = 0.0$ dBm)

Frequency [MHz]	P_{in} **[dBm]**	P_{out1} **[dBm]**	P_{out2} **[dBm]**	**Loss**$_1$ **[dB]**	**Loss**$_2$ **[dB]**
150	0.0	-6.32	-6.19	6.32	6.19
200	0.0	-5.85	-5.89	5.85	5.89
300	0.0	-6.58	-6.59	6.58	6.59

7.4 PART D: Amplifier Characterisation

7.4.1 D1. Gain versus Frequency

Table 6: LNA Output Power and Gain vs. Frequency ($V_{\text{supply}} = 14.47$ V, $P_{\text{in}} = -20.0$ dBm)

Freq. [MHz]	P_{in} **[dBm]**	P_{out} **[dBm]**	**Gain [dB]**	**Freq. [MHz]**	P_{in} **[dBm]**	P_{out} **[dBm]**	**Gain [dB**
100	-20.0	2.40	**22.40**	400	-20.0	1.27	**21.27**
150	-20.0	2.07	**22.07**	450	-20.0	1.21	**21.21**
200	-20.0	1.98	**21.98**	500	-20.0	0.62	**20.62**
250	-20.0	1.68	**21.68**	550	-20.0	0.77	**20.77**
300	-20.0	1.70	**21.70**	600	-20.0	0.46	**20.46**
350	-20.0	1.40	**21.40**				

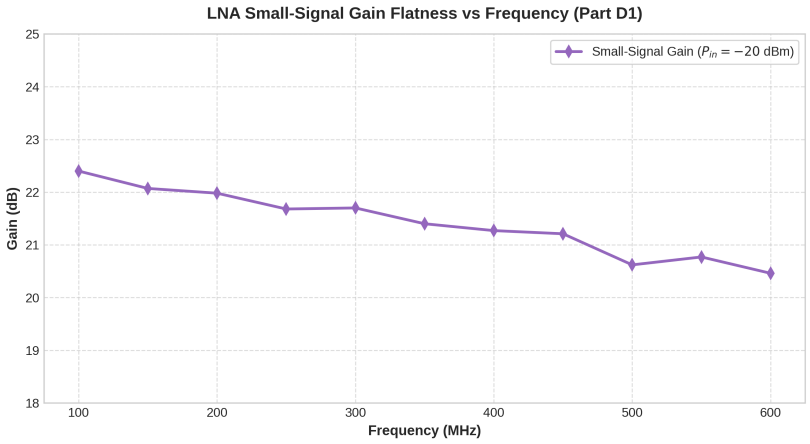


Figure 4: LNA Gain flatness response showing small-signal gain across 100 – 600 MHz.

7.4.2 D2. Harmonics and 1-dB Compression Point

Table 7: LNA Gain Compression Sweep at $f = 100$ MHz

P_{in} [dBm]	P_{out} [dBm]	Gain [dB]	P_{in} [dBm]	P_{out} [dBm]	Gain [dB]
-20.0	2.40	**22.40**	-11.0	11.29	**22.29**
-19.0	3.30	**22.30**	-10.0	12.20	**22.20**
-18.0	4.37	**22.37**	-9.0	13.20	**22.20**
-17.0	5.34	**22.34**	-8.0	14.06	**22.06**
-16.0	6.25	**22.25**	-7.0	14.85	**21.85**
-15.0	7.34	**22.34**	-6.0	15.50	**21.50**
-14.0	8.30	**22.30**	-5.0	16.02	**21.02**
-13.0	9.20	**22.20**	-4.0	16.40	**20.40**
-12.0	10.20	**22.20**			

1-dB Compression Results:

- **Small-Signal Gain (G_0): 22.33 dB**
- **Input P_{1dB} ($P_{\text{in, 1dB}}$): -5.65 dBm**
- **Output P_{1dB} ($P_{\text{out, 1dB}}$): 15.68 dBm**

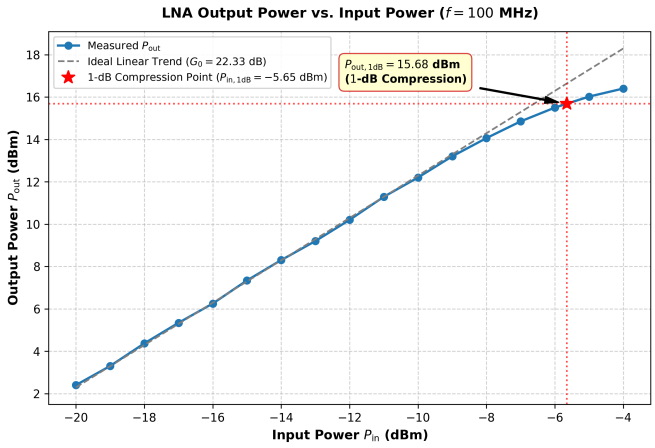


Figure 5: LNA fundamental output power (P_{out}) as a function of input power (P_{in}) at 100 MHz. The curve shows linear amplification at lower powers and saturates as it reaches the Output 1-dB Compression Point at $P_{\text{out, 1dB}} = 15.68$ dBm.

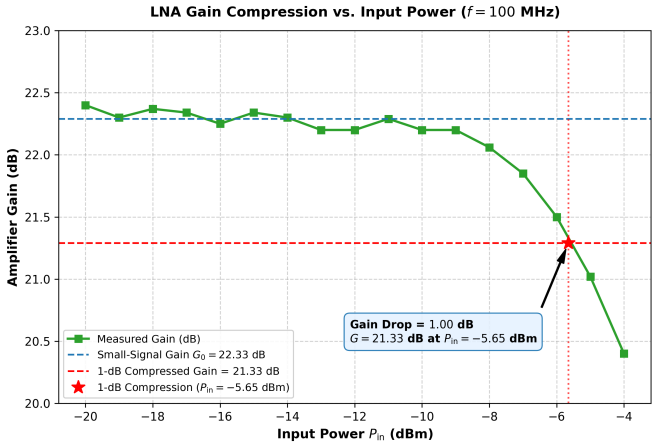


Figure 6: LNA power gain as a function of input power (P_{in}) at 100 MHz. The small-signal gain baseline ($G_0 = 22.33$ dB) drops by 1.0 dB to 21.33 dB at an Input 1-dB Compression Point of $P_{\text{in, 1dB}} = -5.65$ dBm.

7.4.3 D3. Third-Order Intercept Point (TOI)

Table 8: Two-Tone Intermodulation Products Data ($f_1 = 10$ MHz, $f_2 = 13$ MHz)

P_{in}/tone [dBm]	P_{out} [dBm]	P_{IM3} [dBm]	P_{in}/tone [dBm]	P_{out} [dBm]	P_{IM3} [dBm]
-10.0	8.75	-27.62	-4.0	13.83	-9.43
-9.0	9.66	-24.02	-3.0	14.54	-7.32
-8.0	10.78	-20.14	-2.0	15.10	-5.60
-7.0	11.44	-17.34	-1.0	15.70	-4.20
-6.0	12.25	-14.46	0.0	16.13	-3.39
-5.0	13.08	-11.80	+1.0	16.54	-3.01

Extrapolated Intercept Results:

- **Input Third-Order Intercept (IIP3): +5.64 dBm**
- **Output Third-Order Intercept (OIP3): +22.29 dBm**

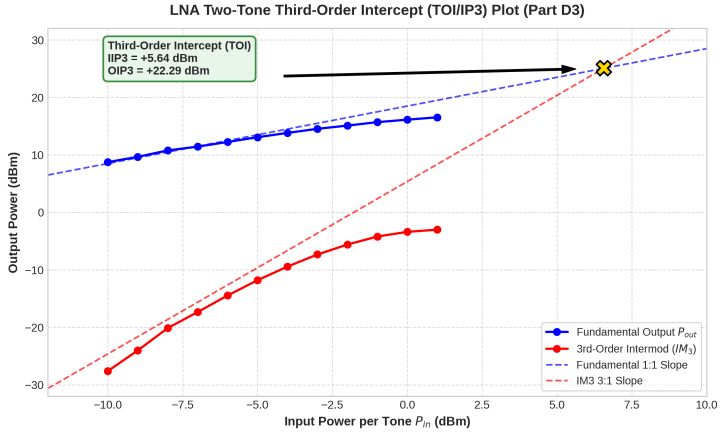


Figure 7: Two-tone third-order intermodulation extrapolation showing the Third-Order Intercept (TOI) point.

7.5 Summary of Principal Measurements

Table 9: Summary of Final Characterization Results

Quantity	**Symbol**	**Measured/Calculated Value**
Low-pass filter cutoff frequency	f_c	550.0 MHz
Low-pass filter stopband edge	f_s	740 MHz
Filter Insertion Loss in Passband	IL	$\approx$ 0.03 to 1.5 dB
Filter Reflection Coefficient	$\lvert\Gamma\rvert$	**0.2788** (at 68 MHz)
Filter Return Loss	RL	**11.09 dB** (at 68 MHz)
Expected Stub Notch Frequency	f_0 (calc)	53.81 MHz
Measured Stub Notch Frequency	f_0 (meas)	57.00 MHz (Peak Attenuation: **39.47 dB**)
Mixer Down-conversion Loss (avg)	Conversion Loss	**12.43 dB**
Power-Splitter Output Powers	$P_{\text{out1}}, P_{\text{out2}}$	Avg: -6.25 dBm / -6.22 dBm
Amplifier Small-Signal Gain	G_0	**22.33 dB**
Amplifier 1-dB Compression Point	P_{1dB}	Input: **-5.65 dBm**, Output: **15.68 dBm**
Third-Order Intercept Point	TOI	IIP3 = +**5.64** dBm, OIP3 = +**22.29** dBm

8 Conclusion

In this experiment we successfully characterized several core RF components used in a standard radio receiver chain: the low-pass filter, the quarter-wave coaxial notch filter, the three-port mixer, and the symmetric power splitter.

- **Low-Pass Filter Characterisation:** By comparing the output power with and without the device, we identified a Low-Pass Filter with an experimental -3 dB cutoff of **550.0 MHz** and a stopband edge from the graph can be read at **740 MHz**.
- **Stub Resonator Performance:** A physical open stub of length 92 cm was shunted across the line, acting as a notch filter. The theoretical quarter-wave resonance was calculated to be **53.81 MHz**, and a very deep notch of **39.47 dB** was observed experimentally at **57.00 MHz**. The +3.19 MHz (+5.9%) frequency offset is physically expected and attributed to the additional electrical length of the Tee-connector adapter.
- **Mixer Operation:** The frequency translation behavior of the mixer was observed at an LO of 30 MHz. For three separate RF inputs, the mixer translated signals to their sum and difference frequencies with a very stable conversion loss averaging **12.43 dB** for down-conversion and **12.50 dB** for up-conversion.
- **Splitter Division:** The symmetric splitter showed balanced power division to Port 1 (average loss 6.25 dB) and Port 2 (average loss 6.22 dB).
- **Low-Noise Amplifier Characterisation (Part D):** The active LNA ($V_{\text{supply}} = 14.47$ V) achieved a flat small-signal gain averaging **22.33 dB** across 100–600 MHz. Power sweep analysis at 100 MHz identified a 1-dB Gain Compression Point at an input power of $P_{\textbf{in, 1dB}} = -5.65$ **dBm** ($P_{\text{out, 1dB}} = 15.68$ dBm).
- **Two-tone intermodulation analysis** (10 MHz, 13 MHz) established an extrapolated Third-Order Intercept point of **IIP3** $= +5.64$ **dBm** and **OIP3** $= +22.29$ **dBm**.
- **Measurement Uncertainties:** Key experimental error sources in this setup include:
  - Impedance of the coaxial cables which increases nonlinearly with frequency.
  - Mismatch reflections at Tee junctions and adapters that generate standing waves .
  - Some components were loose leading to not accurate reading.
  - Higher order Harmonics introduced by the non-linear mixer casued unwanted signal distortion.
  - Heating of equipment adding noise

9 AI Usage Declaration

1. **LaTeX Formatting and Document Structure:** AI was used as a guide to understand LaTeX formatting, helping to structure the document, set up packages, manage margins, and debug compilation errors to ensure a clean and professional presentation.

2. **Polishing and Refining Written Content:** AI was used in correcting the grammar.

3. **Conceptual Clarification:** AI was used as a supplementary tool to clarify theoretical concepts discussed during the experiment, such as how oscilloscope detects the wave or impedance relation with phase velocity.

4. **Plotting and extrapolation**: AI was used to assist in writing Python code for plotting the experimental data. It was used in the extrapolation of the fundamental and third-order intermodulation components and to determine their intersection for estimation of the third-order intercept point (TOI).

5. **Verifying calcualtions :** AI was used in the calculation section to re-verify the calculations.